

Veer Narmad South Gujarat University, Surat



Computer Science and Information Technology Faculty

Syllabus for (SemesterIII) of

B.C.A. (Honours)

302 – Statistical Analysis using 'R'

UNIT 1 & 2 NOTES

by:

Dr. Mohammadali Momin

Unit1: Basic concepts of statistic

- 1.1 Population vs. sample, variables (categorical vs. numerical), datatypes.
- 1.2 Descriptive statistics: measures of central tendency (mean, median, mode),
- 1.3 Measures of dispersion (range, variance, standard deviation)

UNIT 1

Q1: What is the difference between Population and Sample in statistics? Explain with examples.

Answer:

Population:

A population is the entire group that you are interested in studying.

Example: If you want to study the height of all students in a school, then all students in the school form the population.

Sample:

A sample is a small part of the population that you actually study or collect data from.

Example: If you can't measure the height of every student, you may choose 50 students randomly to represent the entire school. This group of 50 students is your sample.

Key Difference:

Population is the whole group, while a sample is a subset (small part) of the population.

Q2: What are variables in statistics? Explain categorical and numerical variables with examples.

Answer:

Variables:

Variables are characteristics or features that can take different values.

Example: If you are studying people, their height, weight, age, or gender are variables.

Categorical Variables:

These are variables that represent categories or labels and don't have any numerical value.

Example: Gender (Male/Female), Eye color (Blue/Brown/Green), Marital Status (Single/Married).

Numerical Variables:

These are variables that represent numbers and can be measured.

Example: Age, Weight, Height.

Key Difference:

Categorical variables represent categories or names, while numerical variables represent measurable numbers.

What is the Meaning of Descriptive Statistics?

Descriptive statistics is a branch of statistics that focuses on describing the characteristics of a sample or a population by using various quantitative methods.

What are the Types of Descriptive Statistics?

Measures of central tendency and measures of dispersion are the two types of descriptive statistics. Apart from this graphs, charts and tables can also be used for a visual representation of data.

What are the Measures of Central Tendency in Descriptive Statistics?

There are three measures of central tendency in descriptive statistics. These are the **mean, median, and mode**.

Q3: What is Mean in statistics? How is it calculated, and what does it tell us about the data?**Answer:**

Mean (also called the average) is a way to find the central value of a data set.

To calculate the mean:

Add all the values in the data set.

Divide the total by the number of values.

Formula:

Mean = (Sum of all values) ÷ (Number of values).

Example:

If your data set is: 5, 7, 9, then the mean is: $(5 + 7 + 9) \div 3 = 21 \div 3 = 7$.

What it tells us:

The mean gives us a rough idea of the "average" value in the data set, though it can be affected by extreme values (outliers).

Examples on Descriptive Statistics

Example 1: Using descriptive statistics, find the mean and mode of the given data.

{1, 4, 6, 1, 8, 15, 18, 1, 5, 1}

Solution: Total number of observations = 10

Sum of observations = $1 + 4 + 6 + 1 + 8 + 15 + 18 + 1 + 5 + 1 = 60$

Mean = $60 / 10 = 6$

Mode = Most frequently occurring observation = 1

Answer: Mean = 6, Mode = 1

Example 2: Find the sample variance of the following data

{7, 11, 15, 18, 36, 43}

Solution: Sample variance formula, $s^2 = \sum \frac{(X_i - \bar{X})^2}{n-1}$

Mean, $\bar{X} = 21.67$, $n = 6$

$$\begin{aligned} s^2 &= [(7 - 21.67)^2 + (11 - 21.67)^2 + (15 - 21.67)^2 + (18 - 21.67)^2 + (36 - 21.67)^2 + (43 - 21.67)^2] / 6 - 1 \\ &= 209.47 \end{aligned}$$

Answer: $s^2 = 209.47$

Example 3: Find the median and the mean deviation about the median for the given data

{9, 10, 12, 16, 17, 17, 18, 20}

Solution: $n = 8$

$$\text{Median} = [(n / 2)^{\text{th}} \text{ term} + ((n / 2) + 1)^{\text{th}} \text{ term}] / 2$$

$$= [(8 / 2)^{\text{th}} \text{ term} + ((8 / 2) + 1)^{\text{th}} \text{ term}] / 2$$

$$= (4^{\text{th}} \text{ term} + 5^{\text{th}} \text{ term}) / 2$$

$$= (16 + 17) / 2 = 16.5$$

$$\text{Mean deviation about median} = \sum_1^n \frac{|X - 16.5|}{n}$$

$$= [|9 - 16.5| + |10 - 16.5| + |12 - 16.5| + |16 - 16.5| + |17 - 16.5| + |17 - 16.5| + |18 - 16.5| + |20 - 16.5|] / 8$$

$$= 3.125$$

Answer: Median = 16.5, Mean deviation about median = 3.125

Q.4: What are Median and Mode? How are they calculated, and why are they useful?

Answer:

Median:

The median is the middle value in a data set when the values are arranged in order (smallest to largest).

Steps to calculate:

Arrange the numbers in order.

If the number of values is odd, the median is the middle number.

If the number of values is even, the median is the average of the two middle numbers.

Example:

If your data set is 3, 5, 8, the median is 5 (middle value).

If your data set is 2, 4, 6, 8, the median is $(4 + 6) \div 2 = 10 \div 2 = 5$.

Mode:

The mode is the value that appears most often in a data set.

Example:

If your data set is 2, 3, 3, 4, 5, the mode is 3 (it appears twice).

Why they are useful:

Median is useful when there are extreme values (outliers) that could affect the mean.

Mode tells us which value occurs the most, useful in understanding frequent trends.

Q5: What is Range in statistics? How is it calculated, and what does it tell us?**Answer:**

Range is the simplest measure of dispersion (spread) in a data set. It shows the difference between the largest and smallest values.

Formula:

Range = Maximum value - Minimum value.

Example:

If the data set is: 3, 7, 10, 15, then the range is $15 - 3 = 12$.

What it tells us:

The range gives us a rough idea of how spread out the data is, but it only looks at the highest and lowest values, so it can be misleading if there are outliers.

Q6: What are Variance and Standard Deviation, and how do they help us understand the spread of data?**Answer:****Variance:**

Variance measures how much the values in a data set differ from the mean.

Steps to calculate:

- > Find the mean of the data.
- > Subtract the mean from each value to find the difference.
- > Square the difference (this removes negatives).
- > Find the average of all these squared differences.

Formula:

Variance = (Sum of squared differences) ÷ (Number of values).

What it tells us:

Variance tells us how spread out the data is. A higher variance means the data is more spread out.

Standard Deviation:

Standard deviation is simply the square root of the variance.

It tells us the average amount by which each data point differs from the mean.

Formula:

Standard Deviation = $\sqrt{\text{Variance}}$.

- Standard deviation gives a more understandable measure of the spread since it's in the same units as the original data.
- A low standard deviation means the data points are close to the mean, while a high standard deviation means they are spread out.

UNIT – 2

Unit2: Data Representation and Sampling technique

- 2.1 Graphical representation of data (histograms, box plots, scatter plots)
- 2.2 Probability theory: basic probability concepts
- 2.3 Probability distributions (binomial, normal distributions)
- 2.4 Sampling techniques: random sampling, stratified sampling,
- 2.5 sampling distributions.
- 2.6 Understanding Bell curve.

1. What are histograms, and how are they used to represent data?

Answer:

A histogram is a graphical representation of data that shows the frequency of data points falling within specified ranges, called bins.

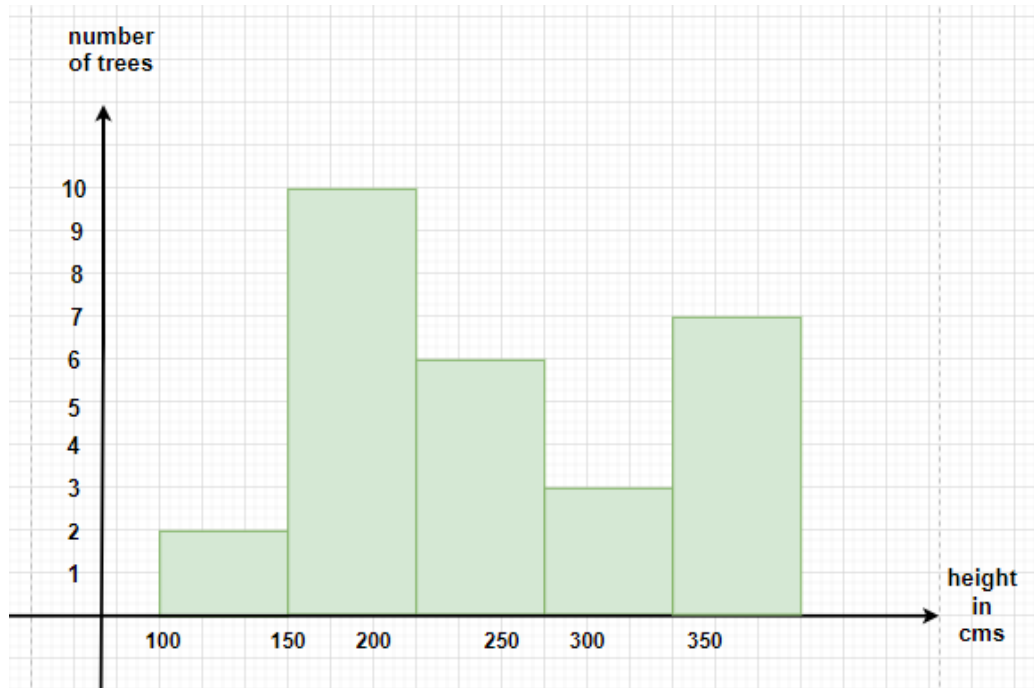
Here's how to create and interpret a histogram:

- 1. Collect Data:** Gather the data you want to represent.
- 2. Choose Bins:** Decide the range of values each bin will cover. For example, if you have exam scores from 0 to 100, you might create bins for 0-10, 11-20, etc.
- 3. Count Frequencies:** Count how many data points fall into each bin.
- 4. Draw the Histogram:** On the x-axis, plot the bins (range of values). On the y-axis, plot the frequency of data points in each bin. Draw bars for each bin where the height represents the frequency.

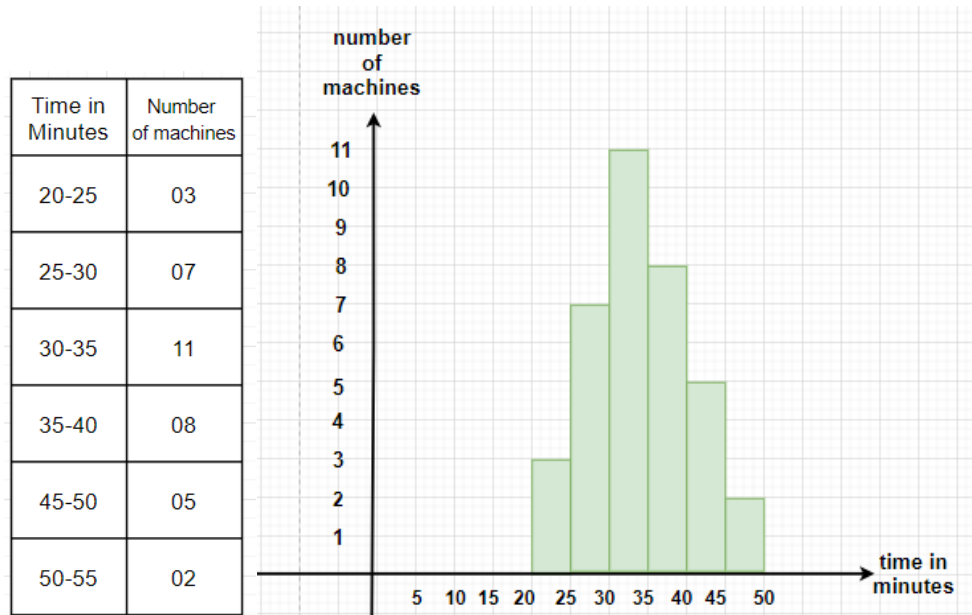
Example1: Suppose you have scores of students on a test: 55, 67, 75, 85, 92. If you choose bins of width 10 (50-60, 60-70, etc.), you can count how many scores fall into each bin and draw bars to visualize this distribution.

Example2: In a Park, there are 28 trees of different heights, the heights can be measured in centimeters and the range of the trees lie between 100-350 cms.

Range of Height of Tree (in cms)	100-150	150-200	200-250	250-300	300-350
Number of Trees	2	10	6	3	7

**Example3:**

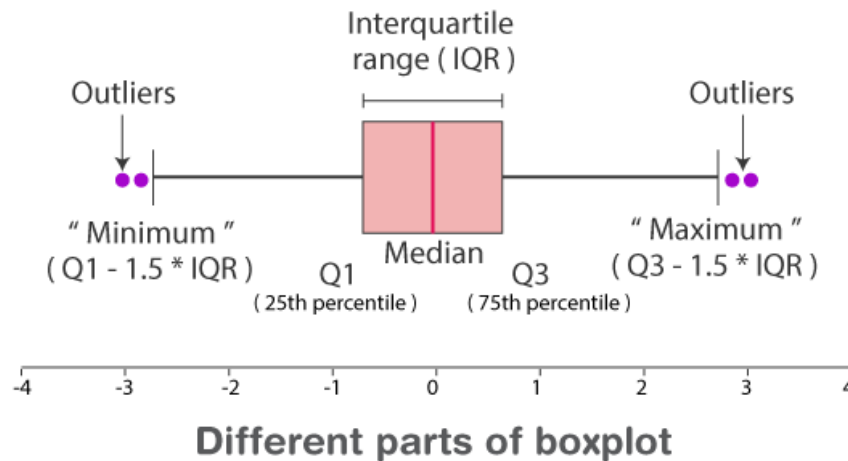
In the belowgiven table, the number of machines working in a factory for their fixed amount of time is shown. Draw the Histogram for the following data.



2. Explain box plots and what information they provide about data.

Answer:

A box plot (or boxandwhisker plot) provides a summary of a dataset's distribution by showing its quartiles and outliers:



- 1. Draw the Box:** Create a box from the first quartile (Q1, the 25th percentile) to the third quartile (Q3, the 75th percentile). The length of the box represents the interquartile range (IQR).
- 2. Draw the Median Line:** Inside the box, draw a line at the median (the 50th percentile).
- 3. Add Whiskers:** Extend lines (whiskers) from the box to the smallest and largest values within 1.5 times the IQR from Q1 and Q3.
- 4. Identify Outliers:** Points outside the whiskers are considered outliers.

Minimum: The minimum value in the given dataset

First Quartile (Q1): The first quartile is the median of the lower half of the data set.

Median: The median is the middle value of the dataset, which divides the given dataset into two equal parts. The median is considered as the second quartile.

Third Quartile (Q3): The third quartile is the median of the upper half of the data.

Maximum: The maximum value in the given dataset.

Apart from these five terms, the other terms used in the box plot are:

Interquartile Range (IQR): The difference between the third quartile and first quartile is known as the interquartile range. (i.e.) $IQR = Q3 - Q1$

Outlier: The data that falls on the far left or right side of the ordered data is tested to be the outliers. Generally, the outliers fall more than the specified distance from the first and third quartile.

Outliers are greater than $Q3 + (1.5 \cdot IQR)$ or less than $Q1 - (1.5 \cdot IQR)$.

Example: A box plot of exam scores might show the median score, the range where most scores fall, and any unusually high or low scores. Find the maximum, minimum, median, first quartile, third quartile for the given data set: 23, 42, 12, 10, 15, 14, 9.

Solution:

Given: 23, 42, 12, 10, 15, 14, 9.

Arrange the given dataset in ascending order.

9, 10, 12, 14, 15, 23, 42

Hence,

Minimum = 9

Maximum = 42

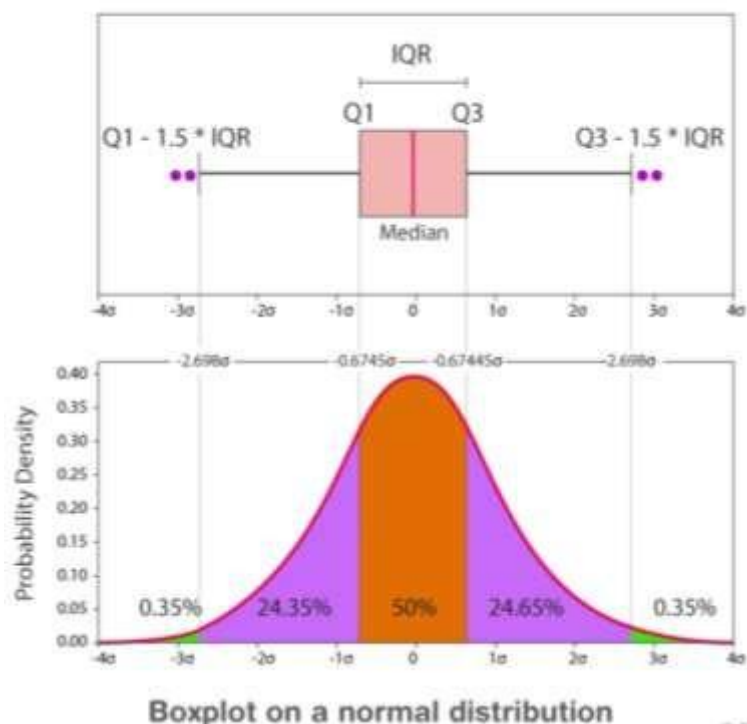
Median = 14

First Quartile = 10 (Middle value of 9, 10, 12 is 10)

Third Quartile = 23 (Middle value of 15, 23, 42 is 23).

Boxplot Distribution:

The box plot distribution will explain how tightly the data is grouped, how the data is skewed, and also about the symmetry of data.



Positively Skewed: If the distance from the median to the maximum is greater than the distance from the median to the minimum, then the box plot is positively skewed.

Negatively Skewed: If the distance from the median to minimum is greater than the distance from the median to the maximum, then the box plot is negatively skewed.

Symmetric: The box plot is said to be symmetric if the median is equidistant from the maximum and minimum values.

3. How do scatter plots help in understanding relationships between two variables?

Answer:

A scatter plot displays the relationship between two quantitative variables:

1. Plot Data Points: Each point on the plot represents a pair of values (one for each variable).
2. Interpret Patterns: Look for trends or patterns:

Positive Correlation: If points trend upwards, as one variable increases, so does the other.

Negative Correlation: If points trend downwards, as one variable increases, the other decreases.

No Correlation: If points are scattered with no discernible pattern, there is no correlation.

Example: A scatter plot showing hours studied versus exam scores can help visualize whether more study hours correlate with higher scores.

4. What are the basic concepts of probability theory?

Answer:

Probability theory involves understanding the likelihood of different outcomes:

1. Probability of an Event: The probability of an event happening is the ratio of the number of favorable outcomes to the total number of possible outcomes. It ranges from 0 (impossible event) to 1 (certain event).

$$\text{Probability} = \frac{\text{Number of Favorable Outcomes}}{\text{Total Number of Possible Outcomes}}$$

2. Complementary Events: The probability that an event does not occur is 1 minus the probability that it does occur.

$$P(\text{Not } A) = 1 - P(A)$$

3. Independent Events: Two events are independent if the occurrence of one does not affect the occurrence of the other.

Example: The probability of rolling a 3 on a fair sixsided die is $\frac{1}{6}$, because there is one favorable outcome and six possible outcomes.

5. Describe the binomial distribution and provide an example.

Answer:

A binomial distribution describes the number of successes in a fixed number of independent trials, each with the same probability of success:

1. Parameters: It is defined by two parameters: (n) (number of trials) and (p) (probability of success).
2. Probability Formula: The probability of getting exactly (k) successes in (n) trials is given by:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

where $\binom{n}{k}$ is the binomial coefficient.

Example: If you flip a coin 3 times, the probability of getting exactly 2 heads (with $p = 0.5$) can be calculated using the binomial formula.

6. What is a normal distribution, and what are its key properties?

Answer:

A normal distribution is a continuous probability distribution that is symmetric around the mean:

- The normal distribution has any positive standard deviation. We know that the mean helps to determine the line of symmetry of a graph, whereas the standard deviation helps to know how far the data are spread out. If the standard deviation is smaller, the data are somewhat close to each other and the graph becomes narrower.
- If the standard deviation is larger, the data are dispersed more, and the graph becomes wider.
- The standard deviations are used to subdivide the area under the normal curve. Each subdivided section defines the percentage of data, which falls into the specific region of a graph.

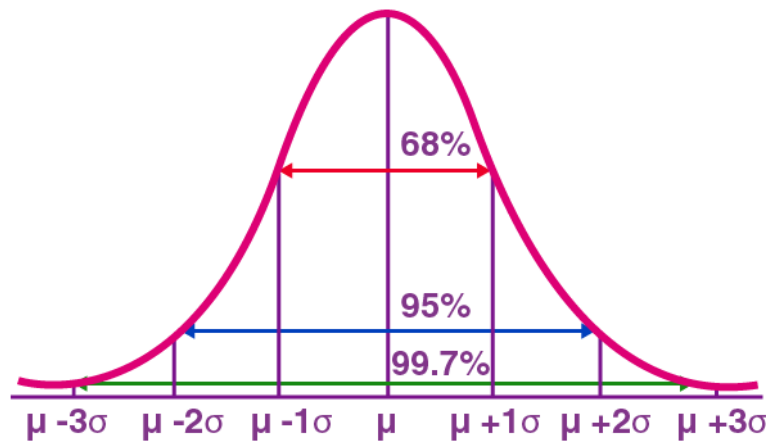
1. Bell Curve: It has a bellshaped curve where the mean, median, and mode are all equal.

2. Parameters: It is defined by two parameters: the mean (μ) and the standard deviation (σ).

3. 68-95-99.7 Rule: About 68% of the data falls within 1 standard deviation of the mean, 95% within 2, and 99.7% within 3.

- Approximately 68% of the data falls within one standard deviation of the mean. (i.e., Between Mean $\pm$ one standard deviation)
- Approximately 95% of the data falls within two standard deviations of the mean. (i.e., Between Mean $\pm$ two standard deviations)
- Approximately 99.7% of the data fall within three standard deviations of the mean. (i.e., Between Mean $\pm$ three standard deviations)

Example: Heights of people typically follow a normal distribution where most people are near the average height, and fewer people are at the extremes.



7. Explain the concept of random sampling and its importance.

Answer:

Random sampling involves selecting a subset of individuals from a population such that each individual has an equal chance of being chosen:

- 1. Method:** Use random methods like drawing names from a hat or using random number generators.
- 2. Purpose:** It ensures that the sample is representative of the population, reducing bias.
- 3. Importance:** It allows for valid generalizations and statistical inferences about the population based on the sample data.

Example: If you want to survey student opinions on campus facilities, randomly selecting students from different classes ensures diverse and unbiased feedback.

8. What is stratified sampling, and how does it differ from random sampling?

Answer:

Stratified sampling involves dividing the population into distinct subgroups (strata) and then performing random sampling within each stratum:

1. Divide Population: Split the population into strata based on characteristics like age, gender, or income.

2. Random Sampling: Perform random sampling within each stratum.

3. Combine Results: Combine the samples from each stratum to form the final sample.

Difference from Random Sampling: Random sampling treats the entire population as one group, while stratified sampling ensures that all important subgroups are represented.

Example: To survey a diverse community, you might stratify by age groups and then randomly sample within each age group to ensure representation across all ages.

9. What is a sampling distribution, and why is it important in statistics?

Answer:

A sampling distribution is the probability distribution of a statistic (like the sample mean) derived from multiple samples of the same size drawn from a population:

1. Construct Sampling Distribution: Take many samples from a population, calculate the statistic for each sample, and create a distribution of these statistics.

2. Key Concept: It helps in understanding the variability of the statistic and making inferences about the population.

Importance: It forms the basis for hypothesis testing and confidence intervals, allowing statisticians to make probabilistic statements about population parameters.

Example: If you repeatedly sample exam scores from a class and calculate the mean for each sample, the distribution of these means will form a sampling distribution of the mean.

10. What is the Bell Curve, and how does it relate to the normal distribution?

Answer:

A bell curve is a type of graph that is used to visualize the distribution of a set of chosen values across a specified group that tend to have a central, normal values, as peak with low and high extremes tapering off relatively symmetrically on either side. Bell curves are visual representations of normal distribution, also called Gaussian distribution.

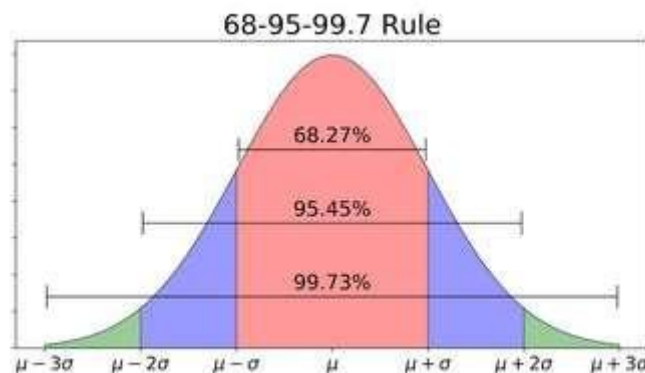
The Bell Curve is a common term for the normal distribution due to its bellshaped curve:

1. Shape: It is symmetric around the mean, with most data points falling close to the mean and fewer points as you move away from the mean.

2. Properties: The curve's shape is determined by the mean and standard deviation. The higher the peak, the smaller the standard deviation.

Relation to Normal Distribution: The Bell Curve is just another name for the normal distribution and describes its characteristic shape and properties.

Example: The distribution of test scores in a large population often forms a Bell Curve, with most students scoring around the average and fewer students scoring very high or very low.



UNIT1 (MCQ)

1. What is a sample?

- a) The entire group you're studying
- b) A small part of the population you study
- c) The result of a survey
- d) A type of numerical variable

Correct Answer: b

2. What is a population in statistics?

- a) A subset of data
- b) A group used for random sampling
- c) The entire group of interest in a study
- d) A specific part of the dataset

Correct Answer: c

3. Which of the following is an example of a categorical variable?

- a) Age
- b) Gender
- c) Weight
- d) Height

Correct Answer: b

4. Which of the following is a numerical variable?

- a) Eye color
- b) Country of birth
- c) Number of children in a family
- d) Marital status

Correct Answer: c

5. What is the key difference between categorical and numerical variables?

- a) Categorical variables are measured in numbers, while numerical variables are not.
- b) Numerical variables represent measurements, while categorical variables represent groups or categories.
- c) Categorical variables change over time, numerical variables don't.
- d) Categorical variables are always qualitative, numerical variables are always quantitative.

Correct Answer: b

6. What is a variable in statistics?

- a) A number that does not change
- b) A characteristic that can take different values
- c) The population being studied
- d) A fixed value

Correct Answer: b

7. Which of the following is an example of a population?

- a) All students in a specific class
- b) 50 randomly chosen students from a school
- c) A survey of 100 people in a city
- d) None of the above

Correct Answer: a

8. If you want to study the income of people in a city, but can only survey 200 people, those 200 people represent the:

- a) Population
- b) Sample
- c) Data type
- d) Variable

Correct Answer: b

9. Which of the following is a nominal categorical variable?

- a) Temperature (in Celsius)
- b) Eye color (blue, green, brown)
- c) Age (in years)
- d) Weight (in kilograms)

Correct Answer: b

10. Which of the following is an ordinal categorical variable?

- a) Height (short, medium, tall)
- b) Gender (male, female)
- c) Weight (50 kg, 60 kg, 70 kg)
- d) Eye color (blue, brown, green)

Correct Answer: a

11. What is the formula for calculating the mean of a dataset?

- a) Sum of all values ÷ number of values
- b) Sum of all values ÷ range of values
- c) Median value ÷ sum of all values
- d) Mode value ÷ sum of all values

Correct Answer: a

12. Which of the following best describes the median?

- a) The most frequent value in a dataset
- b) The middle value in an ordered dataset
- c) The average of all values in a dataset
- d) The difference between the highest and lowest values

Correct Answer: b

13. If the numbers 2, 4, 6, 8, and 10 are in a dataset, what is the mean?

- a) 4

- b) 6
- c) 5
- d) 7

Correct Answer: b

14. In a dataset, if the most frequent value is 12, then this value represents the:

- a) Mean
- b) Mode
- c) Median
- d) Range

Correct Answer: b

15. What is the median of the numbers 1, 2, 3, 4, and 5?

- a) 2
- b) 3
- c) 4
- d) 5

Correct Answer: b

16. Which measure of central tendency is most affected by extreme values (outliers)?

- a) Mean
- b) Median
- c) Mode
- d) Range

Correct Answer: a

17. In which situation would you prefer the median over the mean?

- a) When there are no outliers
- b) When the data is symmetrically distributed
- c) When the data has outliers or is skewed
- d) When calculating the range of data

Correct Answer: c

18. If a dataset has values 3, 3, 4, 5, 8, 9, and 10, what is the mode?

- a) 3
- b) 5
- c) 8
- d) 10

Correct Answer: a

19. Which of the following is true about the mode?

- a) It is always greater than the median.
- b) It can be used for both categorical and numerical data.
- c) It is the same as the median in all datasets.
- d) It is always less than the mean.

Correct Answer: b

20. What is the mean of the numbers 10, 20, 30, 40, and 50?

- a) 30
- b) 25
- c) 35
- d) 40

Correct Answer: a

21. What is the formula for calculating the range of a dataset?

- a) Maximum value - Minimum value
- b) $(\text{Maximum value} + \text{Minimum value}) \div 2$
- c) Sum of all values $\div$ number of values
- d) Maximum value $\times$ Minimum value

Correct Answer: a

22. Which measure of dispersion tells us the average distance of each data point from the mean?

- a) Range
- b) Variance
- c) Standard deviation
- d) Mode

Correct Answer: c

23. What does a low standard deviation indicate about a dataset?

- a) The data points are spread out widely from the mean.
- b) The data points are close to the mean.
- c) The data has many outliers.
- d) The range of the data is high.

Correct Answer: b

24. Which of the following is used to measure how much a data set varies from its mean?

- a) Range
- b) Median
- c) Standard deviation
- d) Mode

Correct Answer: c

25. The range of the dataset 5, 10, 15, 20, and 25 is:

- a) 20
- b) 5
- c) 25
- d) 15

Correct Answer: a

26. What is the variance of a dataset a measure of?

- a) The distance between the largest and smallest values.
- b) How spread out the values are from the mean.
- c) The most common value in the dataset.
- d) The middle value in an ordered list.

Correct Answer: b

27. Which of the following is the square root of the variance?

- a) Mean
- b) Median
- c) Range
- d) Standard deviation

Correct Answer: d

28. A high standard deviation suggests that the data points are:

- a) Very close to the mean
- b) Spread out widely from the mean
- c) All equal
- d) Close to the median

Correct Answer: b

29. If the variance of a data set is 16, what is the standard deviation?

- a) 4
- b) 8
- c) 16
- d) 2

Correct Answer: a

30. Which of the following statements is true about the range?

- a) It is affected by extreme values.
- b) It is a measure of central tendency.
- c) It is always equal to the standard deviation.
- d) It is the square root of the variance.

Correct Answer: a

UNIT – 2 (M.C.Q.)

1. Which graph is best used to display the distribution of a single variable by grouping data into ranges?

- a) Scatter plot
- b) Box plot
- c) Histogram
- d) Bar chart

Correct Answer: c

2. In a box plot, what does the box represent?

- a) The entire range of the data
- b) The interquartile range (IQR)
- c) The mean of the data
- d) The mode of the data

Correct Answer: b

3. What is represented by the line inside the box in a box plot?

- a) Mean
- b) Mode
- c) Median
- d) Maximum value

Correct Answer: c

4. Which type of graph is used to represent the relationship between two numerical variables?

- a) Histogram
- b) Box plot
- c) Scatter plot
- d) Pie chart

Correct Answer: c

5. Which of the following is true about histograms?

- a) They show the relationship between two variables.
- b) They group data into intervals called bins.
- c) They show the spread of data using quartiles.
- d) They are only used for categorical data.

Correct Answer: b

6. A scatter plot is typically used to:

- a) Show the distribution of a single variable.
- b) Compare categories of data.
- c) Show the correlation between two variables.
- d) Display proportions.

Correct Answer: c

7. What is the probability of an event that is certain to happen?

- a) 0
- b) 0.5
- c) 1
- d) 2

Correct Answer: c

8. What is the probability of flipping a fair coin and getting heads?

- a) 0
- b) 0.25
- c) 0.5
- d) 1

Correct Answer: c

9. If two events cannot happen at the same time, they are called:

- a) Independent events
- b) Mutually exclusive events
- c) Complementary events
- d) Conditional events

Correct Answer: b

10. In probability theory, what does "P(A)" represent?

- a) The complement of event A
- b) The probability of event A
- c) The probability of event A not happening
- d) The probability of all events happening

Correct Answer: b

11. What is the range of values for any probability?

- a) 1 to 1
- b) 0 to 1
- c) 0 to 10
- d) 0 to 100

Correct Answer: b

12. If two events are independent, the probability of both events occurring together is:

- a) $P(A) + P(B)$
- b) $P(A) \times P(B)$
- c) $P(A) \div P(B)$
- d) $P(A) - P(B)$

Correct Answer: b

13. Which of the following is a characteristic of a binomial distribution?

- a) Trials are dependent.
- b) The probability of success changes.
- c) There are only two possible outcomes per trial.
- d) Each trial has an infinite number of outcomes.

Correct Answer: c

14. A normal distribution is also known as:

- a) Skewed distribution
- b) Bell curve
- c) Binomial distribution
- d) Flat distribution

Correct Answer: b

15. What is the shape of a normal distribution?

- a) Skewed
- b) Symmetrical
- c) Flat
- d) Irregular

Correct Answer: b

16. In a normal distribution, the mean, median, and mode are:

- a) All different
- b) All equal
- c) Mean is higher than median and mode
- d) Mode is higher than mean and median

Correct Answer: b

17. Which of the following is true for a binomial distribution?

- a) The trials are dependent on each other.
- b) The number of trials is fixed.
- c) The probability of success can be more than 1.
- d) The number of outcomes is unlimited.

Correct Answer: b

18. In a normal distribution, approximately what percentage of data falls within one standard deviation of the mean?

- a) 50%
- b) 68%
- c) 95%
- d) 99.7%

Correct Answer: b

19. What is random sampling?

- a) Dividing the population into groups and sampling from each group
- b) Choosing a sample where every individual has an equal chance of being selected
- c) Choosing only the most convenient individuals
- d) Choosing a sample based on specific characteristics

Correct Answer: b

20. Which of the following is a key characteristic of stratified sampling?

- a) All individuals are equally likely to be selected.
- b) The population is divided into subgroups based on shared characteristics.
- c) The sample is selected randomly from a single group.
- d) The sample is always larger than in random sampling.

Correct Answer: b

21. In which of the following situations would stratified sampling be the most appropriate?

- a) When the population is homogeneous.
- b) When there are distinct subgroups in the population.
- c) When random sampling cannot be done.
- d) When the population is too small.

Correct Answer: b

22. Which of the following is a disadvantage of random sampling?

- a) It is biased.
- b) It is time consuming and expensive.
- c) It requires prior knowledge of the population.
- d) It can only be used with small populations.

Correct Answer: b

23. Which sampling technique ensures that specific subgroups are adequately represented in the sample?

- a) Cluster sampling
- b) Random sampling
- c) Stratified sampling
- d) Systematic sampling

Correct Answer: c

24. What is a sampling distribution?

- a) The distribution of a population
- b) The distribution of sample means or statistics
- c) A distribution of random variables
- d) The distribution of the highest and lowest values in a dataset

Correct Answer: b

25. What happens to the shape of the sampling distribution of the mean as the sample size increases?

- a) It becomes more skewed.
- b) It becomes wider.
- c) It becomes more normal.
- d) It remains unchanged.

Correct Answer: c

26. According to the Central Limit Theorem, the sampling distribution of the mean approaches a normal distribution as:

- a) The sample size decreases
- b) The sample size increases
- c) The population mean decreases
- d) The population variance increases

Correct Answer: b

27. A sampling distribution is made up of:

- a) Individual data points from the population
- b) Data points from multiple samples
- c) The distribution of all possible outcomes
- d) The range of data in one sample

Correct Answer: b

28. Which of the following shapes best represents a normal distribution?

- a) A Ushape
- b) A bell shape
- c) A flat line
- d) A triangle

Correct Answer: b

29. In a bell curve, what percentage of data falls within two standard deviations from the mean?

- a) 68%
- b) 95%
- c) 99.7%
- d) 50%

Correct Answer: b

30. A bell curve is symmetrical, meaning:

- a) The left and right sides are mirror images.
- b) The curve has multiple peaks.
- c) The data is skewed to one side.
- d) The curve is always flat at the center.

Correct Answer: a